

Macroeconomic Theory II, Spring 2024: Homework 5

The assignment is due Friday, 04/25.

Problem 1 In this problem you will compute the equilibrium interest rate in the Aiyagari model. The household solves

$$V(a, \varepsilon) = \max_{c, a'} u(c) + \beta \sum_{\varepsilon' \in E} \pi(\varepsilon' | \varepsilon) V(a', \varepsilon')$$

subject to

$$\begin{aligned} c + a' &= (1 + r)a + w \exp(\varepsilon) \\ a' &\geq -\phi. \end{aligned}$$

The process for ε (in logs) follows an AR(1) process with mean zero, standard deviation of the innovation equal to 0.10 and autocorrelation coefficient equal to 0.90. Discretize it through a 7-state Markov chain by using the Rouwenhorst method. The borrowing limit is $\phi = 0$, i.e., the agent is not allowed to accumulate any unsecured debt. Period utility is given by $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ with $\gamma = 2$ and the discount factor is $\beta = 0.96$. Set the production technology to a CRS Cobb-Douglas function with labor share equal to 0.64 and capital depreciation rate equal to 0.08.

- a) Solve the household problem by using your favorite global solution method.
- b) Plot the aggregate supply of assets and the aggregate capital demand as a function of the interest rate. To compute the stationary distribution use either simulation or approximation of the density function.
- c) Solve for the equilibrium interest rate of this economy.

Problem 2

Time is discrete and indexed by $t \in \{0, 1, \dots, \infty\}$. The economy is populated by a continuum of measure one of infinitely-lived households indexed by i on $[0, 1]$. Preferences for individual i are given by

$$U(c_0^i, c_1^i, \dots) = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i),$$

where $\beta \in (0, 1)$ is the discount factor, $c_t \geq 0$ denotes consumption at time t , $u' > 0$ and $u'' < 0$. Each household i in period t receives a stochastic idiosyncratic endowment of the consumption good $\varepsilon_t^i \in \{\varepsilon_h, \varepsilon_l\}$ which follows a Markov chain where $\pi_{hh}^\varepsilon =$

$\Pr \left\{ \varepsilon_t^i = \varepsilon_h | \varepsilon_{t-1}^i = \varepsilon_h \right\}$, $\pi_{ll}^\varepsilon = \Pr \left\{ \varepsilon_t^i = \varepsilon_l | \varepsilon_{t-1}^i = \varepsilon_l \right\}$, $\pi_{hl}^\varepsilon = 1 - \pi_{hh}^\varepsilon$, and $\pi_{lh}^\varepsilon = 1 - \pi_{ll}^\varepsilon$. Households can trade a risk-free asset a in zero-net supply with return R_t , and they can accumulate debt up to the exogenous borrowing limit $-\bar{b}$.

a) Define a stationary recursive competitive equilibrium (RCE) for this economy, and discuss its existence and uniqueness.

Suppose now that $\bar{b} = 0$, and that $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$.

b) What is so special about this case? Determine the equilibrium risk-free return as a function of the model's parameters and prove that the return is lower than $1/\beta$. Explain how the risk-free rate changes with the ratio $\varepsilon_h/\varepsilon_l$, with the persistence parameter π_{hh} , and with risk aversion γ .

Problem 3

Consider the following variant of the Bewley model. An individual consumer, indexed by $i \in (0, 1)$, chooses consumption c_{it} , hours worked n_{it} and housing h_{it} to solve

$$\max_{c_{it}, n_{it}, h_{it}} \sum_{t=0}^{\infty} \beta^t u(c_{it}, n_{it}, h_{it})$$

subject to

$$P_t c_{it} + Q_t (h_{it+1} - (1 - \delta) h_{it}) + B_{it+1} = W_t e_{it} n_{it} + (1 + i_{t-1}) B_{it},$$

where i_{t-1} is the nominal interest rate in the economy, P_t is the price of the consumption good, Q_t is the price of housing, B_{it+1} is the consumer's holdings of financial assets, W_t is the wage rate per efficiency unit of labor, e_{it} is the consumer's endowment of labor efficiency, and n_{it} is the amount of labor supplied. The consumer's holdings of the financial asset are subject to a borrowing constraint:

$$B_{it+1} \geq -\theta Q_t h_{it+1}.$$

a) List and interpret the first order conditions that characterize the consumer's optimal decision.

In what follows you will derive an alternative formulation of the consumer's problem above, that reduces the number of state variables, and show its equivalence to the original formulation.

b) As written, the consumer's problem is characterized by 3 state variables: holdings of the financial asset, B_{it} , housing wealth, $Q_t h_{it}$ and idiosyncratic labor efficiency, e_{it} . Let $A_{it} = B_{it} + Q_{t-1} h_{it}$ denote the consumer's *net worth*. Rewrite the consumer's budget constraint as a function of its net worth and housing stock.

c) Rescale the budget constraint by P_t , so that $a_{it+1} = A_{it+1}/P_t$, $w_t = W_t/P_t$, $(1+r_t) = (1+i_t)P_t/P_{t+1}$.

d) Assume that the consumer chooses the date t housing stock at the end of period $t-1$, *after* learning the endowment of labor efficiency e_{it} . Write the alternative problem of the consumer recursively.

e) List and interpret the first order conditions in the alternative formulation of the problem.

f) Prove that the sets of first order conditions in the original and the alternative formulation of the problem are identical.

Problem 4

Time is discrete and indexed by $t \in \{0, 1, \dots, \infty\}$. The economy is populated by a continuum of measure one of infinitely-lived households indexed by i on $[0, 1]$. Preferences for individual i are given by

$$U(c_0^i, c_1^i, \dots; h_0^i, h_1^i, \dots) = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i, 1 - h_t^i),$$

where $\beta \in (0, 1)$ is the time discount factor, $c \geq 0$ denotes consumption and $h \geq 0$ hours worked. Household i at time t supplies the optimal amount of hours worked h_t^i to a competitive labor market, taking the hourly wage ω_t as given.

Each household owns a single private firm, i.e. firm i is identified with household i . Each firm produces the final good y (which can be used for consumption or investment) through

$$y_t^i = F(z_t^i, k_t^i, n_t^i),$$

where F is a production technology displaying constant-returns in capital and labor. The variable z_t^i represents a firm-specific productivity shock which follows the Markov process $\pi(z', z) = \Pr\{z_{t+1}^i = z' | z_t^i = z\}$ with $z_{\min} = 0$.

Households can freely hire labor on the market for their firm, but here is no market to trade claims on physical capital, thus households can only invest the capital they own

in their firm. Moreover, $k_t^i \geq 0$. Capital depreciates at rate δ . Let π_t^i denote the profits accruing to individual i from operating her firm at time t .

Households can trade a risk-free bond b in zero-net supply with return R_t and they can accumulate debt up to the “natural borrowing constraint”.

Assume that the economy is in a stationary equilibrium.

- a) What is the natural borrowing constraint faced by each household?
- b) List explicitly the individual state variables and write down the problem of the household in recursive formulation.
- c) Define a Recursive Competitive Equilibrium (RCE) for this economy.
- d) Discuss the existence and uniqueness of the RCE for this economy.

Imagine now that a new financial market opens allowing agents to invest their own k in other households' firms.

- e) What is a simple way to represent the new economy with this additional market? In particular, list what the individual state variables become in this case.